

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                          |
|----------------------|--------------------------|
| United States Patent | 12397697                 |
| Kind Code            | B2                       |
| Date of Patent       | August 26, 2025          |
| Inventor(s)          | Jones; Michael C. et al. |

---

### Liftgate mounting system and method

---

#### Abstract

An apparatus for mounting a liftgate column on a vehicle comprises a plurality of buttons attached to an outer housing of the liftgate column, wherein the plurality of buttons are arranged to project outwardly from a mounting wall of the housing and mate with the plurality of keyholes of the vehicle. Where the vehicle is without corresponding keyholes, a mounting member having corresponding keyholes is disclosed for welding to the vehicle.

---

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| <b>Inventors:</b> | <b>Jones; Michael C. (Torrance, CA), Schauer; David (Omaha, NE)</b> |
| <b>Applicant:</b> | <b>Palfinger USA, LLC (Tiffin, OH)</b>                              |
| <b>Family ID:</b> | <b>1000008777436</b>                                                |
| <b>Assignee:</b>  | <b>PALFINGER USA, LLC (Tiffin, OH)</b>                              |
| <b>Appl. No.:</b> | <b>17/588232</b>                                                    |
| <b>Filed:</b>     | <b>January 29, 2022</b>                                             |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20230242026 A1          | Aug. 03, 2023           |

---

#### Publication Classification

**Int. Cl.:**    **B60P1/44** (20060101)

**U.S. Cl.:**

**CPC**        **B60P1/4485** (20130101); **B60P1/4414** (20130101);

#### Field of Classification Search

**CPC:** B60P (1/4485); B60P (1/4414); B62D (27/06); B62D (27/023)

**USPC:** 296/182.1; 296/186.1; 296/186.3

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name        | U.S. Cl.  | CPC            |
|--------------|-------------|----------------------|-----------|----------------|
| 5685230      | 12/1996     | Tegeler              | 105/397   | B61D<br>17/08  |
| 6322127      | 12/2000     | Masterson            | 296/26.04 | B65D<br>90/14  |
| 9738205      | 12/2016     | Parker               | N/A       | B60P<br>1/4414 |
| 9840181      | 12/2016     | Hambardzumyan et al. | N/A       | N/A            |
| 10625653     | 12/2019     | Hambardzumyan et al. | N/A       | N/A            |
| 10981487     | 12/2020     | Russo                | N/A       | B60P<br>1/4435 |
| 2023/0271540 | 12/2022     | Smolov               | 414/501   | B60P 1/52      |

### FOREIGN PATENT DOCUMENTS

| Patent No.   | Application Date | Country | CPC         |
|--------------|------------------|---------|-------------|
| 102020113965 | 12/2020          | DE      | N/A         |
| 2188027      | 12/1986          | GB      | B60P 1/4485 |

### OTHER PUBLICATIONS

Palfinger Liftgates LLC, ILD Plus Installation Manual & Check-Off Sheet, Mar. 5, 2018, U.S.A. cited by applicant

---

*Primary Examiner:* Daniels; Jason S

*Attorney, Agent or Firm:* Hodgson Russ LLP

---

## Background/Summary

### FIELD OF THE DISCLOSURE

(1) The present disclosure relates to a system and method for mounting a liftgate on a vehicle, for example a truck.

### BACKGROUND OF THE INVENTION

(2) Hydraulically powered liftgates are in widespread use on delivery trucks and other vehicles for assisting operators in loading and unloading cargo. Liftgates commonly have an H-shaped configuration comprising a pair of liftgate columns connected by a traverse member, and a lifting platform arranged between and supported by the columns. The liftgate columns include hydraulic lifting mechanisms operable in unison to raise and lower the liftgate platform between a street-level elevation and a vehicle cargo bed elevation. Liftgates of the type described above may be mounted

to a rear end of the vehicle body as an accessory.

(3) It is known to mount a liftgate to a vehicle body by welding the liftgate columns and traverse member to the corner posts and rear sill at the rear end of the vehicle. Before welding can begin, the mounting process involves preliminary steps including leveling the vehicle, positioning the liftgate with a forklift at a desired position flush with the vehicle corner posts and rear sill, and clamping the liftgate columns to the corner posts of the vehicle. If the rear sill is not flush with the rear surfaces of the corner posts, then additional subframing steps are necessary to weld subframe members to the corner posts to fill the gap and provide flush mounting surfaces on the vehicle. The known mounting process based on welding is disadvantageous because it consumes time, requires trained personnel, and is permanent.

(4) U.S. Pat. Nos. 9,840,181 and 10,625,653 teach a liftgate mounting apparatus and method intended to address the disadvantages mentioned above. The apparatus includes a pair of mounting members configured to be welded to the rear corner posts of a vehicle. Each mounting member includes a plurality of protrusions for mating with corresponding openings in an outer housing of a liftgate column. A securing mechanism is provided within each liftgate column to prevent the liftgate column from shifting and possibly disengaging from the corresponding mounting member. The securing mechanism includes an L-shaped wedge member lodged between an uppermost one of the protrusions and an inner surface of the liftgate column to prevent horizontal shifting. The wedge member is fixed vertically by a clamping nut adjustably set on a vertical bolt extending through a tapped hole in the uppermost protrusion to engage a stop block attached to the inner surface of the liftgate column, whereby the liftgate column is restrained against vertical movement relative to the protrusions on the mounting member. The protrusions and the securing mechanism taught by U.S. Pat. Nos. 9,840,181 and 10,625,653 must compete with mechanical and hydraulic components of the lifting mechanism for free space within the housing of the liftgate column. The securing mechanism requires customized parts such as the L-shaped wedge member and the tapped uppermost protrusion, and is susceptible to loosening over time as the vehicle is used on bumpy streets and/or in environments experiencing wide variations in temperature. If a mounted liftgate is later removed from the vehicle, the mounting members with protrusions remain welded on the vehicle and pose an injury risk to persons walking near the rear end of the vehicle.

#### SUMMARY OF THE DISCLOSURE

(5) The present disclosure describes an apparatus and a method for mounting a liftgate column on a vehicle that overcomes the disadvantages of the current state of the art noted above. According to one embodiment, the vehicle comes equipped with a plurality of keyholes, and the mounting apparatus comprises a plurality of buttons attached to an outer housing of the liftgate column, wherein the plurality of buttons are arranged to project outwardly from a mounting wall of the housing and mate with the plurality of keyholes of the vehicle.

(6) According to another embodiment, the vehicle is not equipped with a plurality of keyholes, and the mounting apparatus further comprises a mounting member configured to be welded to the vehicle. The mounting member includes a plurality of keyholes with which the plurality of buttons attached to the outer housing of the liftgate column are arranged to mate. The mounting member may be part of an H-shaped bracket assembly having two such mounting members extending vertically and a transverse bridge member extending horizontally between the vertical mounting members, such that the two liftgate columns of a liftgate may be respectively mounted on the two mounting members.

(7) A mounting method corresponding to the first embodiment summarized above comprises A) positioning the liftgate column relative to the vehicle such that the plurality of buttons are aligned with the plurality of keyholes; B) moving the liftgate column relative to the vehicle such that the plurality of buttons are received by the plurality of keyholes; and C) lowering the liftgate column relative to the vehicle to mate the plurality of buttons with the plurality of keyholes.

(8) A mounting method corresponding to the second embodiment summarized above comprises A)

providing a mounting member configured to be welded to a vehicle, the mounting member including a plurality of keyholes and a bolt hole; B) positioning the liftgate column relative to the mounting member such that the plurality of buttons are aligned with the plurality of keyholes; C) moving the liftgate column relative to the mounting member such that the plurality of buttons are received by the plurality of keyholes; D) moving the liftgate column relative to the mounting member to mate the plurality of buttons with the plurality of keyholes; E) installing a bolt to extend through the mounting hole in the mounting wall and through the bolt hole of the mounting member; F) threadably engaging the bolt with a nut; and G) welding the mounting member to the vehicle. Steps (A) through (F) may be performed before or after step (G) is performed.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The nature and mode of operation of the present invention will now be more fully described in the following detailed description taken with the accompanying drawing figures, in which:
- (2) FIG. 1 is an exploded perspective view showing a liftgate and a vehicle to which the liftgate is mountable in accordance with a first embodiment of the present disclosure;
- (3) FIG. 1A is an enlarged view of detail region “A” in FIG. 1;
- (4) FIG. 1B is an enlarged view of detail region “B” in FIG. 1;
- (5) FIG. 2 is a front perspective view of a liftgate formed in accordance with the first embodiment of the present disclosure;
- (6) FIG. 3 is a perspective view of an outer housing of a liftgate column having a securement member arranged therein and a plurality of buttons attached thereto in accordance with an aspect of the present disclosure;
- (7) FIG. 4 is another perspective view of the outer housing of the liftgate column showing the securement member arranged therein in accordance with an aspect of the present disclosure;
- (8) FIG. 5 is an enlarged side view showing attachment of one of the plurality of buttons to the outer housing of a liftgate column in accordance with an aspect of the present disclosure;
- (9) FIG. 6 is a side elevational view illustrating a method of mounting a liftgate on a vehicle in accordance with the first embodiment of the present disclosure;
- (10) FIG. 7 is a perspective view illustrating installation of an upper mounting bolt as part of the mounting method;
- (11) FIG. 8 is a perspective view illustrating installation of lower mounting bolts as part of the mounting method;
- (12) FIG. 9 is a cross-sectioned side view showing connection of a liftgate traverse member to a rear sill of the vehicle in accordance with the first embodiment of the present disclosure;
- (13) FIG. 10 is a perspective view showing a bracket assembly of a liftgate mounting system formed in accordance with a second embodiment of the present disclosure, wherein the bracket assembly is configured to be welded to a vehicle;
- (14) FIG. 10A is an enlarged view of detail region “A” in FIG. 10;
- (15) FIG. 10B is an enlarged view of detail region “B” in FIG. 10;
- (16) FIG. 11 is a perspective view showing connection of a transverse bridge member of the bracket assembly to a vertical mounting member of the bracket assembly;
- (17) FIG. 12 is a cross-sectioned side view showing connection of a liftgate traverse member to the bridge member of the bracket assembly in accordance with the second embodiment of the present disclosure;
- (18) FIG. 13 is a perspective view of a pre-assembled liftgate and mounting bracket assembly according to an alternative version of the second embodiment;
- (19) FIG. 13A is an enlarged view of detail region “A” in FIG. 13; and

(20) FIG. 13B is an enlarged view of detail region “B” in FIG. 13.

## DETAILED DESCRIPTION OF THE INVENTION

(21) FIG. 1 shows a liftgate **10** and a vehicle **100** to which liftgate **10** is removably mountable in accordance with a first embodiment of the present disclosure.

(22) Liftgate **10** generally comprises a pair of liftgate columns **12** substantially mirroring each other about a vertical centerline of liftgate **10**, a traverse member **14** connecting the liftgate columns **12**, and a foldable liftgate platform **16** supported between liftgate columns **12** for movement up and down relative to columns **12**. Each liftgate column **12** includes an outer housing **18** and a hydraulically-powered lifting mechanism **20** within outer housing **18**. The lifting mechanism **20** of each liftgate column **12** is coupled to platform **14**. The respective lifting mechanisms **20** are operable in unison to raise and lower platform **14** along columns **12**.

(23) Vehicle **100** is depicted, by way of example, as a delivery truck having a rear door **102** permitting access to a cargo bed of the truck, a pair of rear corner posts **104** extending vertically on opposite sides of rear door **102**, and a rear sill **106** extending transversely across the vehicle between rear corner posts **104** at the level of the cargo bed.

(24) In the first embodiment of the present disclosure shown in FIGS. 1-9, the rear corner posts **104** and rear sill **106** are integral original parts of vehicle **100** that are configured for mounting liftgate **10** on vehicle **100**, and liftgate **10** is configured for removable mounting on rear corner posts **104** and rear sill **106** as will be described in detail below. Rear corner posts **104** and rear sill **106** may be formed of a steel or stainless steel alloy, and surface treated (e.g., galvanized) by the manufacturer of the vehicle **100** as part of the vehicle manufacturing process.

(25) Each rear corner post **104** may include a plurality of keyholes **108** and one or more bolt holes **110** formed through a rearwardly facing wall **105** of the corner post **104**. Rear sill **106** may include a plurality of bolt holes **110** through a rearwardly facing wall **107** of rear sill **106**. Rear corner posts **104** and rear sill **106** may be formed of metal channel to provide open interior space within the posts **104** and sill **106**. Vehicle **100** may further include a plurality of nuts **112**, shown for example in FIGS. 1A and 1B, that are fixed to interior surfaces of rearwardly facing walls **105** and **107** in respective alignment and association with bolt holes **110**. The plurality of keyholes **108** of each rear corner post **104** may be vertically spaced from one another along the corner post **104**. Each keyhole **108** may include a narrow portion **108A** and a wide portion **108B** above the narrow portion **108A**.

(26) Each liftgate column **12** includes a plurality of buttons **30** attached to outer housing **18** of the liftgate column. Buttons **30** are arranged to project outwardly from a mounting wall **22** of outer housing **18** to respectively mate with the plurality of keyholes **108** of vehicle **100** when mounting wall **22** is positioned against rearwardly facing wall **205** of the associated corner post **104**. The plurality of buttons **30** of each liftgate column **12** may be vertically spaced from one another along the column **12**. Each button **30** may include a stem **30A** extending from mounting wall **22** and a head **30B** at a distal end of the stem **30A**, wherein the stem **30A** is at least as narrow as the narrow portion **108A** of the corresponding keyhole **108**, and the head **30B** is wider than the narrow portion **108A** of the keyhole but fits within the wide portion **108B** of the keyhole. Buttons **30** may be case hardened so the buttons resist wear and avoid fracture.

(27) As best seen in FIG. 5, each button **30** may be attached to mounting wall **22** by a corresponding threaded fastener **32** arranged to extend through an axial hole **31** in the button **30** and through a corresponding fastener hole **24** through mounting wall **22** to the interior of outer housing **18**, wherein the threaded fastener **32** mates in threaded engagement with a threaded hole **25** aligned with fastener hole **24** and fixed relative to outer housing **18**.

(28) In an aspect of the present disclosure, threaded holes **25** corresponding to fastener holes **24** and buttons **30** may be provided on a prefabricated securement member **40** fixedly attached, for example by welding and/or fasteners, to the interior surface of mounting wall **22**. Securement member **40** may be formed from a strip **42** of metal plate drilled with holes **44** corresponding to fastener holes **24** in mounting wall **22**, and having a plurality of nuts **46** welded to a rear surface of

strip **42** in alignment with holes **44**.

(29) Each liftgate column **12** may include one or more mounting holes **26** through mounting wall **22**. Each mounting hole **26** may be arranged to align with a corresponding bolt hole **110** and fixed nut **112** on vehicle **100** when buttons **30** are mated with keyholes **108** on vehicle **100**. For example, in the present embodiment, a mounting hole **26** is located above the plurality of buttons **30**. While only one mounting hole **26** is depicted for each lift column **12** in FIGS. **1** and **2**, one or more additional mounting holes **26** may be provided on each lift column **12**, for example at a lower portion of each lift column **12** as shown in FIG. **8**. Also as illustrated by FIG. **8**, mounting holes **26** may be formed through a side wall **23** of an extension of outer housing **18** for alignment with bolt holes (not shown) through a side wall of rear corner post **104**.

(30) Traverse member **14** of liftgate **10** may include a plurality of mounting holes **28** arranged at spaced intervals along the traverse member to align with bolt holes **110** in rearwardly facing wall **107** of rear sill **106** and with fixed nuts **112** on the interior surface of rearwardly facing wall **107**.

(31) A method of mounting liftgate **10** on vehicle **100** in accordance with the first embodiment will now be described. As shown in FIG. **6**, the vehicle **100** may be parked and leveled such that rear corner posts **104** extend vertically, and liftgate **10** may be supported on a forklift **F** such that liftgate columns **12** extend vertically and face the rear corner posts **104**. Liftgate **10** may be moved relative to vehicle **100**, for example by operating forklift **F**, to position each liftgate column **12** relative to vehicle **100** such that the plurality of buttons **30** on each liftgate column **12** are aligned with the plurality of keyholes **108** in the corresponding rear corner post **104**. More specifically, the enlarged head **30B** of each button **30** is aligned with the wide portion **108B** of the corresponding keyhole **108**.

(32) Next, liftgate **10** including liftgate columns **12** is moved relative to vehicle **100**, for example by advancing forklift **F** toward the rear of vehicle **100**, such that the plurality of buttons **30** are received by the plurality of keyholes **108**. As may be understood, the enlarged head **30B** of each button **30** is inserted into the wide portion **108B** of the corresponding keyhole **108**.

(33) Once the buttons **30** projecting from each liftgate column **12** are received by keyholes **108** in the associated rear corner post **104**, liftgate **10** including liftgate columns **12** may be lowered relative to vehicle **100**, for example by lowering the carriage and forks of forklift **F**, to mate the buttons **30** with the keyholes **108**. In particular, liftgate **10** is lowered such that the narrow stem **30A** of each button **30** passes through the narrow portion **108A** of the keyhole. At this stage, liftgate **10** is supportably mounted on vehicle **100** and is not removable except by raising liftgate **10** relative to vehicle **100** and withdrawing liftgate **10** away from the vehicle.

(34) The method may include further steps for securing each liftgate column **12** to its associated rear corner post **104** to prevent the liftgate columns **12** from moving upward relative to vehicle **100** and potentially becoming disengaged from vehicle **100** when the vehicle **100** is being driven on a bumpy roadway. In the illustrated embodiment, a respective bolt **34** may be installed by arranging the bolt to extend through mounting hole **26** in mounting wall **22** and through the bolt hole **110** in vehicle **100**, and threadably engaging the bolt **34** with the fixed nut **112** associated with the bolt hole **110**. By way of non-limiting example, and as shown in FIG. **7**, a top cover **19** removably fastened to outer housing **18** may be temporarily removed, and bolt **34** may be inserted into the interior of outer housing **18**, positioned through mounting hole **26** and bolt hole **110**, and tightened in threaded engagement with fixed nut **112**. If additional mounting holes **26** are provided as shown in FIG. **8**, and rear corner post **104** is configured with corresponding bolt holes **110** and fixed nuts **112**, then additional bolts **34** may be installed to further secure liftgate column **12** to rear corner post **104**. Bolts, nuts, and other hardware mentioned herein may be Grade A hardware.

(35) The method may include further steps for securing traverse member **14** of liftgate **10** to rear sill **106** of vehicle **100**. As illustrated in FIG. **9**, a respective bolt **34** may be arranged to extend through each mounting hole **28** in traverse member **14** and through the corresponding bolt hole **110** in rear sill **106** of vehicle **100** to threadably engage with the fixed nut **112** associated with the bolt

hole **110**. Traverse member **14** may be a segment of rectangular tubing having a plurality of openings **29** through a bottom wall of the tubing at locations corresponding to mounting holes **28** to facilitate installation of bolts **34** through mounting holes **28**.

(36) Once liftgate **10** is securely mounted on vehicle **100** as described above, the carriage and forks of forklift **F** may be lowered further to relinquish support of liftgate **10** to vehicle **100**, and forklift **F** may be moved away from vehicle **100**.

(37) Advantageously, the first embodiment described above enables liftgate **10** to be mounted on vehicle **100** without any welding operations. As a result, the task of mounting liftgate **10** on vehicle **100** requires less time and less skill on the part of the labor team.

(38) In accordance with the first embodiment, liftgate **10** may be removed from vehicle **100** quickly and easily by reversing the mounting steps described above. For example, if liftgate **10** becomes damaged, it can be removed and serviced, and a replacement liftgate may be mounted on vehicle **100** while the damaged liftgate is repaired. Or, if a new liftgate is desired, it is a simple matter to swap a previously mounted liftgate for a new liftgate.

(39) In the first embodiment described above, the rear corner posts **104** and rear sill **106** of vehicle **100** are configured for mounting liftgate **10** by virtue of keyholes **108**, bolt holes **110**, and fixed nuts **112**. In a second embodiment of the present disclosure described with reference to FIGS. **10-13B**, the rear corner posts **104** and rear sill **106** of vehicle **100** are not configured for mounting liftgate **10** on vehicle **100**. Thus, in accordance with the second embodiment, a bracket assembly **200** may be provided for welding onto vehicle **100** to configure the vehicle for removable mounting of liftgate **10**.

(40) Bracket assembly **200** may include first and second vertical mounting members **204** and a transverse bridge member **206** connecting the first and second vertical mounting members **204**, wherein each of the first and second vertical mounting members **204** comprises a plurality of keyholes **208**. As illustrated in FIG. **11**, transverse bridge member **206** may connect to each vertical mounting member **204** by way of a pair of end tabs **214** respectively protruding from opposite ends of ends of transverse bridge member **206** for receipt by corresponding slots **215** facing sidewalls of vertical mounting members **204**, only one end tab **214** and slot **215** being visible in FIG. **11**.

Vertical mounting members **204** and transverse bridge member **206** may be formed of a suitable material such as a steel alloy or stainless steel alloy. Surface treatment of the members **204**, **206** may be selected based at least in part on the expected type of vehicle **100** and expected end use environmental conditions. For example, members **204**, **206** may be galvanized, painted, and/or treated with other types of protective coatings. With regard to protective surface treatments, extra attention may be paid to the weld seams by which members **204** and **206** are attached to vehicle **100**, as described below, subsequent to attachment.

(41) Each of the vertical mounting members **204** may further comprise one or more bolt holes **210** formed through a rearwardly facing wall **205** of the vertical mounting member **204**. Transverse bridge member **206** may include a plurality of bolt holes **210** through a rearwardly facing wall **207** of the transverse bridge member. Vertical mounting members **204** and transverse bridge member **206** may be formed of metal channel to provide open interior space within the vertical mounting members **204** and bridge member **206**. Bracket assembly **200** may further include a plurality of nuts **212** fixed to interior surfaces of rearwardly facing walls **205** and **207** in respective alignment and association with bolt holes **210**, as shown in FIGS. **10A** and **10B**. The plurality of keyholes **208** of each vertical mounting member **204** may be vertically spaced from one another along the vertical mounting member **204**. Similar to keyholes **108** of the first embodiment, each keyhole **208** may include a narrow portion **208A** and a wide portion **208B** above the narrow portion **208A**.

(42) As may be understood, the keyholes **208**, bolt holes **210**, and fixed nuts **212** of bracket assembly **200** serve the same functions as keyholes **108**, bolt holes **110**, and fixed nuts **112** of the first embodiment, and thus may have the same or a similar configuration as keyholes **108**, bolt holes **110**, and fixed nuts **112** of the first embodiment for mating with features of liftgate **10** as

described above.

(43) Bracket assembly **200** of the second embodiment may be attached to vehicle **100** separately from liftgate **10**, as illustrated by FIG. **10**. For example, bracket assembly **200** may be welded to the rear of vehicle **100**. Each vertical mounting member **204** may be welded to a respective rear corner post **104** of vehicle **100** by a series of intermittent welds on each side of vertical mounting member **204**. By way of non-limiting example, the welds may be ¼-inch fillet welds, each two-inches in length, spaced at eighteen-inch intervals along the length of vertical mounting member **204**. Transverse bridge member **206** may be welded to rear sill **106** of vehicle **100** by a series of intermittent welds along the top and bottom edges of transverse bridge member **206**. For example, if vehicle **100** has a width of ninety-six inches, then the welds may be ¼-inch fillet welds, each two-inches in length, spaced at twenty-inch intervals along the length of transverse bridge member **206**. As another example, if vehicle **100** has a width of one-hundred two inches, then the spacing interval of the welds may be adjusted to twenty-two inches.

(44) In the example described above, it is presumed that the rearwardly facing surfaces of rear corner posts **104** and rear sill **106** are flush with one another, i.e., all surfaces lie in the same plane. Some vehicles may have configurations with irregular shaped sills, wherein the rearwardly facing surface of rear sill **106** is not flush with the rearwardly facing surfaces of corner posts **104**. These situations may be remedied with a process referred to as subframing and/or flushing the sill, whereby a subframe is built up on rear corner posts **104** using correct size 3/16 inch or greater wall tubing. The welding installation procedure for mounting assembly **200** does not change.

(45) Once bracket assembly **200** is fixedly attached to vehicle **100**, liftgate **10** may be mounted on vehicle **100** and removed from vehicle **100** as described above in reference to the first embodiment.

(46) In an alternative version of the second embodiment, bracket assembly **200** and liftgate **10** may be attached to vehicle **100** as a preassembled unit, as illustrated by FIG. **13**. In the alternative version, liftgate **10** is preinstalled on bracket assembly **200** prior to welding bracket assembly **200** on vehicle **100**. Liftgate **10** is preinstalled on bracket assembly **200** by mating buttons **30** on liftgate columns **12** with keyholes **208** on vertical mounting members **204**. Liftgate **10** may be secured to bracket assembly **200** by arranging respective bolts **34** to extend through corresponding mounting holes **26** in liftgate column mounting walls **22** and through corresponding bolt holes **210** in vertical mounting members **204**, and threadably engaging the bolt **34** with the fixed nut **212** associated with the bolt hole **210**, in a manner analogous to that disclosed for securing liftgate columns **12** to rear corner posts **104** in the first embodiment. Additionally, liftgate **10** may be secured to bracket assembly **200** by arranging respective bolts **34** to extend through mounting holes **28** in traverse member **14** and through corresponding bolt holes **210** in transverse bridge member **206** of bracket assembly **200** to threadably engage with fixed nuts **212** associated with the bolt holes **210**, in a manner analogous to that disclosed for securing traverse member **14** to rear sill **106** in the first embodiment.

(47) Prior to welding bracket assembly **200** with attached liftgate **10** on vehicle **100**, preliminary steps may be carried out to help ensure the bracket assembly is properly positioned against vehicle **100** when the bracket assembly is welded to the vehicle. Vehicle **100** may be levelled by parking the vehicle on level ground, and adjusting tire pressures if necessary, such that rearwardly facing surfaces of rear corner posts **104** and rear sill **106** are vertical. The liftgate **10** and bracket assembly **200** may be safely attached to and supported by a forklift, and the forklift may be operated to position vertical mounting members **204** and transverse bridge member **206** flush against rear corner posts **104** and rear sill **106**, respectively. After positioning bracket assembly **200** against vehicle **100**, centering may be checked by measuring the distance from the outer edge of each vertical mounting member **204** to the outer edges of the corresponding rear corner post **104** at various locations along the vertical mounting members **204** to ensure the distances are equal. Bracket assembly may be measured to confirm that vertical mounting members **204** are square and parallel, for example by using a framing square to verify vertical mounting members **204** are square



at 90° to transverse bridge member **206** and to vehicle rear sill **106**. Once proper positioning is confirmed, liftgate **10** and bracket assembly **200** may be securely clamped flush against vehicle **100**, for example by using four “F” style clamps, two at the top and bottom of one liftgate column **12**, and two at the top and bottom of the other liftgate column **12**. Finally, precautionary measures may be taken so that vertical mounting members **204** and liftgate columns **12** do not “toe-out” during welding. For example, flat metal bar or round stock may be added at the corner junctions to act as braces to minimize pull from welding.

(48) Once the preliminary steps have been carried out, bracket assembly **200** may be welded to vehicle **100**, for example in the manner described above with reference to FIG. **10**. Following completion of the welding operation, the forklift, clamps, and braces may be removed.

(49) As may be understood, liftgate **10** may be removed from vehicle **100** and remounted on vehicle **100** as described above.

(50) As used herein, the term “fixed nut” encompasses not only a nut that has been fixed to a structural member, for example by welding the nut to the member, but also a hole that has been drilled and tapped in a structural member, and a hole that has been drilled in a structural member and fitted with a threaded insert.

(51) While the present disclosure describes exemplary embodiments, the detailed description is not intended to limit the scope of the appended claims to the particular embodiments set forth. The claims are intended to cover such alternatives, modifications and equivalents of the described embodiments as may be included within the scope of the claims.

## Claims

1. An apparatus for mounting a liftgate on a vehicle, wherein the liftgate comprises a first liftgate column, a second liftgate column, and a traverse member connecting the first and second liftgate columns, each of the first and second liftgate columns comprising an outer housing having a mounting wall, the apparatus comprising: a bracket assembly configured to be welded to the vehicle, the bracket assembly including a first vertical mounting member, a second vertical mounting member, and a transverse bridge member connecting the first and second vertical mounting members, wherein each of the first and second vertical mounting members includes a plurality of keyholes; a first plurality of buttons attached to the outer housing of the first liftgate column, the first plurality of buttons being arranged to project outwardly from the mounting wall of the first liftgate column and mate with the plurality of keyholes of the first vertical mounting member; and a second plurality of buttons attached to the outer housing of the second liftgate column, the second plurality of buttons being arranged to project outwardly from the mounting wall of the second liftgate column and mate with the plurality of keyholes of the second vertical mounting member; wherein the first vertical mounting member further comprises a bolt hole and a fixed nut associated with the bolt hole, the first liftgate column further comprises a mounting hole through the mounting wall thereof, and the apparatus further comprises a first bolt configured to extend through the mounting hole in the mounting wall of the first liftgate column and through the bolt hole in the first vertical mounting member to threadably engage with the fixed nut of the first vertical mounting member; wherein the second vertical mounting member further comprises a bolt hole and a fixed nut associated with the bolt hole, the second liftgate column further comprises a mounting hole through the mounting wall thereof, and the apparatus further comprises a second bolt configured to extend through the mounting hole in the mounting wall of the second liftgate column and through the bolt hole in the second vertical mounting member to threadably engage with the fixed nut of the second vertical mounting member wherein the transverse bridge member of the bracket assembly comprises a plurality of auxiliary bolt holes and a plurality of auxiliary fixed nuts respectively associated with the plurality of auxiliary bolt holes in the transverse bridge member; wherein the traverse member of the liftgate comprises a plurality of auxiliary mounting

holes through a front wall thereof; and wherein the apparatus further comprises a plurality of auxiliary bolts configured to extend respectively through the plurality of auxiliary mounting holes in the front wall of the traverse member and through the plurality of auxiliary bolt holes in the transverse bridge member to threadably engage with the plurality of auxiliary fixed nuts of the transverse bridge member.

2. The apparatus according to claim 1, wherein: each of the plurality of keyholes of the first vertical mounting member and each of the plurality of keyholes of the second vertical mounting member includes a narrow portion and a wide portion above the narrow portion; each of the first plurality of buttons includes a stem extending from the mounting wall of the first liftgate column and a head at a distal end of the stem, wherein each of the first plurality of buttons mates with a corresponding one of the plurality of keyholes of the first vertical mounting member by inserting the head of the button into the wide portion of the corresponding keyhole and lowering the liftgate relative to the mounting member such that the stem of the button passes through the narrow portion of the corresponding keyhole, wherein the head of the button is wider than the narrow portion of the corresponding keyhole; and each of the second plurality of buttons includes a stem extending from the mounting wall of the second liftgate column and a head at a distal end of the stem, wherein each of the second plurality of buttons mates with a corresponding one of the plurality of keyholes of the second vertical mounting member by inserting the head of the button into the wide portion of the corresponding keyhole and lowering the liftgate relative to the mounting member such that the stem of the button passes through the narrow portion of the corresponding keyhole, wherein the head of the button is wider than the narrow portion of the corresponding keyhole.

3. A method for mounting a liftgate column on a vehicle, wherein the liftgate column comprises an outer housing having a mounting wall including a mounting hole therethrough, and a plurality of buttons attached to the outer housing and arranged to project outwardly from the mounting wall, the method comprising: A) providing a mounting member configured to be welded to a vehicle, the mounting member including a plurality of keyholes and a bolt hole; B) positioning the liftgate column relative to the mounting member such that the plurality of buttons are aligned with the plurality of keyholes; C) moving the liftgate column relative to the mounting member such that the plurality of buttons are received by the plurality of keyholes; D) moving the liftgate column relative to the mounting member to mate the plurality of buttons with the plurality of keyholes; E) installing a bolt to extend through the mounting hole in the mounting wall and through the bolt hole of the mounting member; F) threadably engaging the bolt with a nut; and G) welding the mounting member to the vehicle.

4. The method according to claim 3, wherein the liftgate column is a first liftgate column of a liftgate assembly having a second liftgate column opposite the first liftgate column, the method further comprising: providing another mounting member configured to be welded to the vehicle, the another mounting member including a plurality of keyholes and a bolt hole; performing steps (B) through (G) with respect to the second liftgate column and the another mounting member.

5. The method according to claim 3, wherein steps (A) through (F) are performed before performing step (G).

6. The method according to claim 3, wherein step (G) is performed before performing any of steps (B) through (F).

---